

Kuwait University

2021/2022

Spring course



Calculus B

Second

Sections : 11.4 part (b/b) – 11.5

11.4 The Comparison Tests.

11.5 Alternating Series.

(11)



Limit Comparison Test (L.C.T)



Suppose that

$$\sum a_n$$

and

$$\sum b_n$$

are series with positive terms

if $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c > 0$

Then

either both series **converge** or **diverge**



استثناءات النظرية



Suppose

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$$

and

$$\sum b_n \text{ is convergent}$$

Then

$$\sum a_n \text{ is convergent}$$

Suppose

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$$

and

$$\sum b_n \text{ is divergent}$$

Then

$$\sum a_n \text{ is divergent}$$

طريقة الحل



(1) استنتاج السيكونس القوية b_n

(2) أدرس السيرييس لها

Convergent - divergent

(3) أخذ اللمت للسؤال الأصلي ضرب مقلوب القوية

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c > 0$$

طريقة استنتاج القوية



1) a^n

2) $n^{\dots}$

$$b_n = \frac{\text{أقوى حد بالبسط}}{\text{أقوى حد بالمقام}}$$

$$\sum \frac{n^4 + 5^n}{n^2 + 2^n} \rightarrow b_n = \frac{5^n}{2^n}$$

$$\sum \frac{\sqrt{n}}{n^5 + 2n^2} \rightarrow b_n = \frac{n^{\frac{1}{2}}}{n^5}$$

$$\sum \frac{3^n + 4^n}{2^n + 6^n} \rightarrow b_n = \frac{4^n}{6^n}$$

$$\sum \frac{(n-3)e^n}{\sqrt{n^2-1}} \rightarrow b_n = \frac{ne^n}{\sqrt{n^2}}$$

$$\sum \frac{2n^2 + n}{3^n(n^2-5)} \rightarrow b_n = \frac{n^2}{3^n n^2}$$

$$\sum \frac{n \tan\left(\frac{1}{n}\right)}{n^2 + 3} \rightarrow b_n = \frac{n \frac{1}{n}}{n^2}$$

$$\sum \sin\left(\frac{1}{n}\right) \rightarrow b_n = \frac{1}{n}$$

$$\sum \frac{n + \ln(n)}{n^2 + 5} \rightarrow b_n = \frac{n}{n^2}$$

Determine whether the series is convergent or divergence

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n+1}}$$

$\rightarrow \sum_{n=1}^{\infty} a_n$

Let $b_n = \frac{1}{\sqrt[3]{n}} = \frac{1}{n^{\frac{1}{3}}}$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{3}}} \quad P = \frac{1}{3} < 1$$

is divergent (P-series)

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt[3]{n+1}} \quad \times \quad n^{\frac{1}{3}}$$

$$\lim_{n \rightarrow \infty} \frac{n^{\frac{1}{3}}}{\sqrt[3]{n(1+\frac{1}{n})}} = \lim_{n \rightarrow \infty} \frac{\cancel{n^{\frac{1}{3}}}}{\sqrt[3]{n} \sqrt[3]{1+\frac{1}{n}}}$$

$$= \frac{1}{\sqrt[3]{1+0}} = \frac{1}{1} = 1 \quad \neq 0$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n+1}}$ is divergent (L.C.T)

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt[3]{n+1}} = \frac{1}{\infty} = 0$$



فشل
N.T.T

$$\frac{1}{\sqrt[3]{n+1}} \sim \frac{1}{\sqrt[3]{n}}$$

$$\sum \frac{1}{n^{\frac{1}{3}}} \quad P = \frac{1}{3} < 1$$

Divergent (P-series)



فشل
B.C.T

Determine whether the series is convergent or divergence

$$\sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{1}{\sqrt{n}}$$

Let $b_n = \frac{1}{n} \cdot \frac{1}{\sqrt{n}} = \frac{1}{n^{\frac{3}{2}}}$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}} \quad P = \frac{3}{2} > 1$$

is convergent (P-series)

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sin \frac{1}{\sqrt{n}} \cdot n^{\frac{3}{2}} = \lim_{n \rightarrow \infty} n^{\frac{3}{2}} \sin \frac{1}{\sqrt{n}}$$

$$\lim_{n \rightarrow \infty} n^{\frac{1}{2}} \sin \frac{1}{\sqrt{n}}$$

$$\lim_{n \rightarrow \infty} n^{\frac{1}{2}} \cdot \frac{\sin \frac{1}{\sqrt{n}}}{\frac{1}{\sqrt{n}}} = 1 > 0$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{1}{\sqrt{n}}$ is convergent (L.C.T)

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sin \frac{1}{\sqrt{n}} = \frac{1}{\infty} \sin \frac{1}{\infty} = 0 \sin 0 = 0$$



فشل
N.T.T

$$-1 \leq \sin \frac{1}{\sqrt{n}} \leq 1$$

$$\frac{-1}{n} \leq \frac{1}{n} \sin \frac{1}{\sqrt{n}} \leq \frac{1}{n} \rightarrow \text{Divergent}$$



فشل
B.C.T

~~G.T.H.P.I~~

N.B.L

Determine whether the series is convergent or divergence

$$\sum_{n=1}^{\infty} \frac{\sin\left(\frac{1}{n}\right)}{\sqrt[3]{n+1}}$$

Let $b_n = \frac{\frac{1}{n}}{\sqrt[3]{n}} = \frac{1}{n^{\frac{1}{3}} n} = \frac{1}{n^{\frac{4}{3}}}$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^{\frac{4}{3}}} \quad p = \frac{4}{3} > 1$$

is convergent (P-series)

$$\lim_{n \rightarrow \infty} \frac{\sin\left(\frac{1}{n}\right)}{\sqrt[3]{n+1}} \quad * n^{\frac{4}{3}}$$

$$\lim_{n \rightarrow \infty} \frac{\sin\left(\frac{1}{n}\right)}{\sqrt[3]{n(1+\frac{1}{n})}} \quad n^{\frac{4}{3}} = \lim_{n \rightarrow \infty} \frac{\sin\left(\frac{1}{n}\right)}{\sqrt[3]{n} \sqrt[3]{1+\frac{1}{n}}} \quad n^{\frac{4}{3}}$$

$$\lim_{n \rightarrow \infty} \frac{\sin\left(\frac{1}{n}\right)}{\sqrt[3]{1+\frac{1}{n}}} \quad * n$$

$$\lim_{n \rightarrow \infty} \frac{\frac{\sin\left(\frac{1}{n}\right)}{\frac{1}{n}}}{\sqrt[3]{1+\frac{1}{n}}} \quad \frac{1}{n} \quad * n$$

$$= \frac{1}{\sqrt[3]{1+0}} = \frac{1}{\sqrt[3]{1}} = 1 > 0$$

$\therefore \sum_{n=1}^{\infty} \frac{\sin\left(\frac{1}{n}\right)}{\sqrt[3]{n+1}}$ is convergent (L.C.T)



Text book questions



Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{1}{n^2 + 1} \quad \xrightarrow{\text{red arrow}} a_n$$

Let $b_n = \frac{1}{n^2}$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^2} \quad p=2 > 1$$

Convergent (P-series)

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n^2 + 1} \cdot n^2 = \lim_{n \rightarrow \infty} \frac{n^2}{n^2 \left[1 + \frac{1}{n^2} \right]} = \frac{1}{1+0} = 1 \neq 0$$

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent (L.C.T)

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{n^3}{n^4 + 4} \quad \xrightarrow{\text{red arrow}} a_n$$

Let $b_n = \frac{n^3}{n^4} = \frac{1}{n}$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n} \quad \text{divergent (Harmonic)}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n^3}{n^4 + 4} \cdot n$$

$$\lim_{n \rightarrow \infty} \frac{n^4}{n^4 \left[1 + \frac{4}{n^4} \right]} = \frac{1}{1+0} = 1 \neq 0$$

$\therefore \sum_{n=1}^{\infty} a_n$ is divergent (L.C.T)

Determine whether the series is convergent or divergence

$$\sum_{n=1}^{\infty} \frac{1}{2^n - 1} \rightarrow \sum_{n=1}^{\infty} a_n$$

Let $b_n = \frac{1}{2^n} = \left(\frac{1}{2}\right)^n$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n \rightarrow r = \frac{1}{2}$$

$|r| < 1$

is convergent (geometric Series)

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$\lim_{n \rightarrow \infty} \frac{1}{2^n - 1}$$

$$\lim_{n \rightarrow \infty} \frac{2^n}{2^n - 1}$$

$\div 2^n$

$$\lim_{n \rightarrow \infty} \frac{1}{1 - \left(\frac{1}{2}\right)^n} = \frac{1}{1 - 0} = 1 > 0$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{1}{2^n - 1}$ is convergent (L.C.T)

Determine whether the series is convergent or divergence

$$\sum_{n=0}^{\infty} \frac{3^{2n}}{10^n - 2n^{10}}$$

$$\sum_{n=0}^{\infty} \frac{9^n}{10^n - 2n^{10}}$$

Let $b_n = \frac{9^n}{10^n} = \left(\frac{9}{10}\right)^n$

$$\sum_{n=0}^{\infty} b_n = \sum_{n=0}^{\infty} \left(\frac{9}{10}\right)^n \rightarrow r = \frac{9}{10}$$

$\therefore |r| < 1$

is convergent (geometric Series)

$$\lim_{n \rightarrow \infty} \frac{9^n}{10^n - 2n^{10}} \quad * \frac{10^n}{9^n}$$

$$\lim_{n \rightarrow \infty} \frac{10^n}{10^n - 2n^{10}}$$

$\div 10^n$

$$\lim_{n \rightarrow \infty} \frac{1}{1 - \frac{2n^{10}}{10^n}} = \frac{1}{1 - 0} = 1 \neq 0$$

$$\downarrow \lim_{n \rightarrow \infty} \frac{2n^{10}}{10^n}$$

Let $f(x) = \frac{2x^{10}}{10^x}$

$$\lim_{x \rightarrow \infty} \frac{2x^{10}}{10^x} = 0 \quad (\text{using L'Hop Rule 10 times})$$

$$\Rightarrow \therefore \sum_{n=0}^{\infty} \frac{9^n}{10^n - 2n^{10}} \text{ is convergent (L.C.T.)}$$

Determine whether the series

$$\sum_{n=1}^{\infty} \frac{2n^3 - n + 1}{3n^4 + 2n^3 - n - 1} \rightarrow a_n$$

is convergent or divergent

Dec 2016

Let $b_n = \frac{n^3}{n^4} = \frac{1}{n}$

$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n}$ is divergent (Harmonic)

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$\lim_{n \rightarrow \infty} \frac{2n^3 - n + 1}{3n^4 + 2n^3 - n - 1}$$

$$\lim_{n \rightarrow \infty} \frac{n^3 \left[2 - \frac{1}{n^2} + \frac{1}{n^3} \right]}{n^4 \left[3 + \frac{2}{n} - \frac{1}{n^3} - \frac{1}{n^4} \right]}$$

$$= \frac{2 - 0 + 0}{3 + 0 - 0 - 0} = \frac{2}{3} \neq 0$$

$\therefore \sum_{n=1}^{\infty} a_n$ is divergent (L.C.T)

Determine whether the series is convergent or divergence

$$\sum_{n=1}^{\infty} \frac{\tan\left(\frac{1}{n}\right)}{n+1}$$

Let $b_n = \frac{\frac{1}{n}}{n} = \frac{1}{n^2}$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^2} \quad P=2 > 1$$

is convergent (P-series)

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$\lim_{n \rightarrow \infty} \frac{\tan\left(\frac{1}{n}\right)}{n+1} \quad * n^2$$

$$\lim_{n \rightarrow \infty} \frac{\tan\left(\frac{1}{n}\right)}{n\left(1+\frac{1}{n}\right)} \quad \cdot n^2$$

$$\lim_{n \rightarrow \infty} \frac{\tan\left(\frac{1}{n}\right)}{1+\frac{1}{n}} \quad n$$

$$\lim_{n \rightarrow \infty} \frac{\tan\left(\frac{1}{n}\right)}{1+\frac{1}{n}} = \frac{1}{1+0} = 1 > 0$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{\tan\left(\frac{1}{n}\right)}{n+1}$ is convergent (L.C.T)



Text book questions



Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^2 + 1}} \rightarrow a_n$$

Let $b_n = \frac{1}{\sqrt{n^2}} = \frac{1}{n}$

$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n}$ divergent (Harmonic)

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n^2 + 1}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n^2(1 + \frac{1}{n^2})}} = \frac{1}{\sqrt{1+0}} = 1 \neq 0$$

$\therefore \sum_{n=1}^{\infty} a_n$ is divergent (L.C.T)

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$$

Let $b_n = \frac{1}{n}$

$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n}$ divergent (Harmonic)

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$\lim_{n \rightarrow \infty} \sin\left(\frac{1}{n}\right) * n = \lim_{n \rightarrow \infty} \frac{\sin\left(\frac{1}{n}\right)}{\frac{1}{n}} = 1 \neq 0$$

$\therefore \sum_{n=1}^{\infty} a_n$ is divergent (L.C.T)

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{(\sqrt{n} + 1) \tan^{-1} n}{n^2 - n + 2} \rightarrow \sum_{n=1}^{\infty} a_n$$

Let $b_n = \frac{\sqrt{n}}{n^2} = \frac{1}{n^{\frac{3}{2}}}$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}} \quad p = \frac{3}{2} > 1$$

is convergent (p-series)

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$\lim_{n \rightarrow \infty} \frac{(\sqrt{n} + 1) \tan^{-1} n}{n^2 - n + 2} \quad * \quad n^{\frac{3}{2}}$$

$$\lim_{n \rightarrow \infty} \frac{\sqrt{n} (1 + \frac{1}{\sqrt{n}}) \tan^{-1} n}{n^2 (1 - \frac{1}{n} + \frac{2}{n^2})} \quad n^{\frac{3}{2}}$$

$$= \frac{(1 + 0) \frac{\pi}{2}}{1 - 0 + 0} = \frac{\pi}{2} > 0$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} a_n$ is convergent (L.C.T)



Text book questions



Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{9^n}{3 + 10^n}$$

Let $b_n = \frac{9^n}{10^n}$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{9^n}{10^n} = \sum_{n=1}^{\infty} \left(\frac{9}{10}\right)^n$$

$r = \frac{9}{10}$
 $|r| < 1$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$\lim_{n \rightarrow \infty} \frac{9^n}{3 + 10^n} \div \frac{9^n}{10^n} = \lim_{n \rightarrow \infty} \frac{9^n}{3(9^n) + 9^n} \div (9^n)$$

$$\lim_{n \rightarrow \infty} \frac{1}{3\left(\frac{9}{10}\right)^n + 1} = \frac{1}{0 + 1} = 1 > 0$$

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent (L.C.T)

$\therefore \sum_{n=1}^{\infty} \left(\frac{9}{10}\right)^n$ is convergent (Geometric)

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{\arctan n}{n^{1.2}}$$

$$\sum_{n=1}^{\infty} \frac{\tan^{-1} n}{n^{1.2}}$$

Let $b_n = \frac{1}{n^{1.2}}$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^{1.2}}$$

$p = 1.2 > 1$
Convergent (P-series)

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$\lim_{n \rightarrow \infty} \frac{\tan^{-1} n}{n^{1.2}} = \frac{\pi}{2} > 0$$

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent (L.C.T)

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{2n-1}{\sqrt{n^3}} \longrightarrow \sum_{n=1}^{\infty} a_n$$

Let $b_n = \frac{n}{\sqrt{n^3}} = \frac{n}{n^{\frac{3}{2}}} = \frac{1}{n^{\frac{1}{2}}}$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{2}}} \quad p = \frac{1}{2} < 1$$

is divergent (P-series)

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$\lim_{n \rightarrow \infty} \frac{2n-1}{\sqrt{n^3}}$$

$$\lim_{n \rightarrow \infty} \frac{2n-1}{n}$$

$$\lim_{n \rightarrow \infty} \frac{n[2 - \frac{1}{n}]}{n}$$

$$n^{\frac{1}{2}} - \frac{1}{n^{\frac{1}{2}}} = n^{-\frac{2}{2}} = n^{-1} = \frac{1}{n}$$

$$= 2 - 0 = 2 > 0$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} a_n$ is divergent (L.C.T)

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{n^3 + 3^n}{n^2 + 4^n}$$

Let $b_n = \frac{3^n}{4^n} = \left(\frac{3}{4}\right)^n$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \left(\frac{3}{4}\right)^n \quad r = \frac{3}{4}$$

$|r| < 1$

is convergent (geometric series)

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$\lim_{n \rightarrow \infty} \frac{n^3 + 3^n}{n^2 + 4^n}$$

$$\lim_{n \rightarrow \infty} \frac{4^n n^3 + 12^n}{3^n n^2 + 12^n}$$

$$\lim_{n \rightarrow \infty} \frac{\left(\frac{4}{12}\right)^n n^3 + 1}{\left(\frac{3}{12}\right)^n n^2 + 1} = \lim_{n \rightarrow \infty} \frac{\left(\frac{1}{3}\right)^n n^3 + 1}{\left(\frac{1}{4}\right)^n n^2 + 1}$$

$$\lim_{n \rightarrow \infty} \frac{\frac{n^3}{3^n} + 1}{\frac{n^2}{4^n} + 1} = \frac{0+1}{0+1} = 1 > 0$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} a_n$ is convergent (L.C.T)

$\lim_{n \rightarrow \infty} \frac{n^3}{3^n}$
Let $f(x) = \frac{x^3}{3^x}$
 $\lim_{x \rightarrow \infty} \frac{x^3}{3^x}$

$\lim_{x \rightarrow \infty} \frac{3x^2}{3^x \ln 3}$
 $\lim_{x \rightarrow \infty} \frac{6x}{3^x (\ln 3)^2}$
 $\lim_{x \rightarrow \infty} \frac{6}{3^x (\ln 3)^3} = \frac{6}{\infty} = 0$

$\lim_{n \rightarrow \infty} \frac{n^2}{4^n}$
Let $f(x) = \frac{x^2}{4^x}$

$\lim_{x \rightarrow \infty} \frac{x^2}{4^x}$
 $\lim_{x \rightarrow \infty} \frac{2x}{4^x \ln 4}$
 $\lim_{x \rightarrow \infty} \frac{2}{4^x (\ln 4)^2} = \frac{2}{\infty} = 0$

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{2n^2 + 3n}{\sqrt{4 + n^5}} \rightarrow \sum_{n=1}^{\infty} a_n$$

Let $b_n = \frac{n^2}{\sqrt{n^5}} = \frac{n^2}{n^{\frac{5}{2}}} = \frac{1}{n^{\frac{1}{2}}}$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{2}}} \quad p = \frac{1}{2} < 1$$

is divergent (p-series)

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$\lim_{n \rightarrow \infty} \frac{2n^2 + 3n}{\sqrt{4 + n^5}} \quad n^{\frac{1}{2}}$$

$$\lim_{n \rightarrow \infty} \frac{2n^2 + 3n}{\sqrt{n^5 \left(\frac{4}{n^5} + 1 \right)}} \quad n^{\frac{1}{2}}$$

$$\lim_{n \rightarrow \infty} \frac{2n^2 + 3n}{\sqrt{n^5} \sqrt{\frac{4}{n^5} + 1}} \quad n^{\frac{1}{2}}$$

$$\lim_{n \rightarrow \infty} \frac{n^2 \left[2 + \frac{3}{n} \right]}{n^{\frac{5}{2}} \sqrt{\frac{4}{n^5} + 1}} \quad n^{\frac{1}{2}} = \frac{2+0}{\sqrt{0+1}} = 2 > 0$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} a_n$ is divergent (L.C.T)

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{8n^2 - 7}{e^n(n+1)^2} \rightarrow \sum_{n=1}^{\infty} a_n$$

Let $b_n = \frac{n^2}{e^n n^2} = \frac{1}{e^n} = \left(\frac{1}{e}\right)^n$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \left(\frac{1}{e}\right)^n \rightarrow r = \frac{1}{e}$$

$|r| < 1$

is convergent (geometric series)

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$\lim_{n \rightarrow \infty} \frac{8n^2 - 7}{e^n(n+1)^2}$$

$$\lim_{n \rightarrow \infty} \frac{n^2 \left[8 - \frac{7}{n^2}\right]}{\left(n\left(1 + \frac{1}{n}\right)\right)^2}$$

$$\lim_{n \rightarrow \infty} \frac{n^2 \left[8 - \frac{7}{n^2}\right]}{n^2 \left(1 + \frac{1}{n}\right)^2} = \frac{8-0}{(1+0)^2} = 8 > 0$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} a_n$ is convergent (L.C.T)

Determine whether the series is convergent or divergent

$$\sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{n}{n^2+1} \right) \quad \rightarrow \sum_{n=2}^{\infty} a_n$$

$$\sum_{n=2}^{\infty} \left(\frac{n^2+1}{(n-1)(n^2+1)} - \frac{n(n-1)}{(n^2+1)(n-1)} \right)$$

$$\sum_{n=2}^{\infty} \frac{\cancel{n^2+1} - \cancel{n^2} + n}{(n-1)(n^2+1)}$$

$$\sum_{n=2}^{\infty} \frac{1+n}{(n-1)(n^2+1)}$$

Let $b_n = \frac{n}{n^2} = \frac{1}{n^2}$

$$\sum_{n=2}^{\infty} \frac{1}{n^2} \quad p=2 > 1$$

is convergent (p-series)

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{1+n}{(n-1)(n^2+1)} \quad * n^2$$

$$\lim_{n \rightarrow \infty} \frac{\cancel{n} \left(1 + \frac{1}{n}\right)}{\cancel{n} \left(1 - \frac{1}{n}\right) \cancel{n^2} \left(1 + \frac{1}{n^2}\right)} \quad \cancel{n^2}$$

$$= \frac{(1+0)}{(1-0)(1+0)} = \frac{1}{(1)(1)} = 1 > 0$$

$\Rightarrow \therefore \sum_{n=2}^{\infty} a_n$ is convergent (L.C.T)



Nice Question



Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{\sin(n^2)}{n(2n-1)} \quad \rightarrow \quad \sum_{n=1}^{\infty} a_n$$

$$-1 \leq \sin(n^2) \leq 1$$

$$\div n(2n-1)$$

$$\frac{-1}{n(2n-1)} \leq \frac{\sin(n^2)}{n(2n-1)} \leq \frac{1}{n(2n-1)}$$

$$\sum_{n=1}^{\infty} \frac{1}{n(2n-1)}$$

Let $b_n = \frac{1}{n \cdot n} = \frac{1}{n^2}$
 $\sum_{n=1}^{\infty} \frac{1}{n^2} \quad p=2 > 1$

is convergent (p-series)

$$\lim_{n \rightarrow \infty} \frac{1}{n(2n-1)} \quad * n^2$$

$$\lim_{n \rightarrow \infty} \frac{1}{n \cdot n (2 - \frac{1}{n})} \quad * n^2$$

$$= \frac{1}{2-0} = \frac{1}{2} > 0$$

$$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{1}{n(2n-1)} \text{ is convergent (L.C.T)}$$

$$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{\sin(n^2)}{n(2n-1)} \text{ is convergent (B.C.T)}$$



Nice question !!



Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{\sin(n) \sin\left(\frac{1}{n}\right)}{n+6}$$

$$-1 \leq \sin(n) \leq 1$$

$$* \frac{\sin\left(\frac{1}{n}\right)}{n+6}$$

$$-\frac{\sin\left(\frac{1}{n}\right)}{n+6} \leq \frac{\sin(n) \sin\left(\frac{1}{n}\right)}{n+6} \leq \frac{\sin\left(\frac{1}{n}\right)}{n+6} \leq \frac{\sin\left(\frac{1}{n}\right)}{n}$$

$$\sum_{n=1}^{\infty} \frac{\sin\left(\frac{1}{n}\right)}{n}$$

$$\text{Let } b_n = \frac{\frac{1}{n}}{n} = \frac{1}{n^2}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \quad P=2 > 1$$

is convergent (P-series)

$$\lim_{n \rightarrow \infty} \frac{\sin\left(\frac{1}{n}\right)}{n} * n^2$$

$$\lim_{n \rightarrow \infty} \sin\left(\frac{1}{n}\right) * n$$

$$\lim_{n \rightarrow \infty} \frac{\sin\left(\frac{1}{n}\right)}{\frac{1}{n}} = \frac{1}{1+0} = 1 > 0$$

$$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{\sin\left(\frac{1}{n}\right)}{n} \text{ is convergent (L.C.T)}$$

$$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{\sin(n) \sin\left(\frac{1}{n}\right)}{n+6} \text{ is convergent (B.C.T)}$$

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{e^{\frac{1}{n}}}{\sqrt{n} \cos^2 n}$$

$0 < \cos^2 n < 1$
* $\sqrt{n}$

$0 < \sqrt{n} \cos^2 n < \sqrt{n}$

* $e^{\frac{1}{n}}$

$$\frac{1}{\sqrt{n} \cos^2 n} \gg \frac{1}{\sqrt{n}}$$

$$\frac{e^{\frac{1}{n}}}{\sqrt{n} \cos^2 n} \gg \frac{e^{\frac{1}{n}}}{\sqrt{n}}$$



~~G.T.H.P.~~
~~N.a.B.L~~

$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

Let $b_n = \frac{1}{\sqrt{n}}$

$\sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{2}}}$

$p = \frac{1}{2} < 1$

is Divergent (P-series)

$\lim_{n \rightarrow \infty} \frac{e^{\frac{1}{n}}}{\sqrt{n}}$

* $\sqrt{n} = e^0 = 170$

$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{e^{\frac{1}{n}}}{\sqrt{n}}$ is divergent (L.C.T)

$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{e^{\frac{1}{n}}}{\sqrt{n} \cos^2 n}$ is divergent (B.C.T)

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{2n + \ln(n)}{n^2 + 1} \rightarrow \sum_{n=1}^{\infty} a_n$$

$$+2n \quad \ln n < n \quad \forall n > 0$$

$$2n + \ln n < 3n$$

$$\div (n^2 + 1)$$

$$\frac{2n + \ln n}{n^2 + 1}$$

$$< \frac{3n}{n^2 + 1}$$

$$< \frac{3n}{n^2}$$

Div.

فصل

~~G.T.H.P.I~~

~~N.a.B.L~~

Let $b_n = \frac{n}{n^2} = \frac{1}{n}$
 $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent (Harmonic)

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$\lim_{n \rightarrow \infty} \frac{2n + \ln(n)}{n^2 + 1}$$

$$\lim_{n \rightarrow \infty} \frac{n \left[2 + \frac{\ln(n)}{n} \right]}{n^2 \left[1 + \frac{1}{n^2} \right]}$$

$$= \frac{2+0}{1+0} = 2 > 0$$

Let $f(x) = \frac{\ln x}{x}$

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x}$$

L'Hop $\lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1}$

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} a_n$ is divergent (L.C.T)

قراءة



تفلسف !!



Determine whether the series is convergent or divergent

$$\sum_{n=2}^{\infty} \frac{\ln(n)}{n^2} \rightarrow \sum_{n=2}^{\infty} a_n$$

Let $b_n = \frac{1}{n^{\frac{3}{2}}}$

$$\sum_{n=2}^{\infty} b_n = \sum_{n=2}^{\infty} \frac{1}{n^{\frac{3}{2}}}$$

$$p = \frac{3}{2} > 1$$

is convergent (P-series)

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n^2}$$

$$* n^{\frac{3}{2}}$$

$$= \lim_{n \rightarrow \infty} \frac{\ln n}{n^{\frac{1}{2}}} \quad \frac{\infty}{\infty}$$

Let $f(x) = \frac{\ln x}{x^{\frac{1}{2}}}$

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x^{\frac{1}{2}}}$$

L'HOP

$$\lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{1}{2} x^{-\frac{1}{2}}} = \lim_{x \rightarrow \infty} \frac{2x^{\frac{1}{2}}}{x}$$

$$\lim_{x \rightarrow \infty} \frac{2}{x^{\frac{1}{2}}} = \frac{2}{\infty} = 0$$

$\therefore \sum_{n=2}^{\infty} b_n$ is convergent and $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$

$\Rightarrow \therefore \sum_{n=2}^{\infty} a_n$ is convergent (L.C.T)

Determine whether the following series is convergent or divergent

$$\frac{1}{2^2} + \frac{2^2}{3^3} + \frac{3^3}{4^4} + \dots$$

Next class

$$\sum_{n=1}^{\infty} \frac{n^n}{(n+1)^{n+1}} = \sum_{n=1}^{\infty} \frac{n^n}{(n+1)^n (n+1)^1} \rightarrow \sum_{n=1}^{\infty} a_n$$

Let $b_n = \frac{n^n}{n^n n} = \frac{1}{n}$

$\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent (Harmonic)

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n^n}{(n+1)^n (n+1)} \quad * n$$

$$\lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n \frac{n}{n+1}$$

$$\lim_{n \rightarrow \infty} \left(\frac{n+1-1}{n+1} \right)^n \frac{n}{n+1}$$

$$\lim_{n \rightarrow \infty} \left(\frac{n+1}{n+1} + \frac{-1}{n+1} \right)^n \frac{n}{n(1+\frac{1}{n})}$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{-1}{n+1} \right)^n \frac{1}{1+\frac{1}{n}}$$

$$= e^{-1} \frac{1}{1+0} = e^{-1} = \frac{1}{e} \neq 0$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} a_n$ is divergent (L.C.T)

Determine whether the sequence

$$\{s_n\}_{n=1}^{\infty} = \sin(1) + \sin\left(\frac{1}{2}\right) + \sin\left(\frac{1}{3}\right) + \cdots + \sin\left(\frac{1}{n}\right)$$

is convergent or divergent.

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$$\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right) \rightarrow a_n$$

$$\text{Let } b_n = \frac{1}{n}$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n} \text{ is divergent (Harmonic Series)}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$\lim_{n \rightarrow \infty} \sin\left(\frac{1}{n}\right) \cdot n$$

$$\lim_{n \rightarrow \infty} \frac{\sin\left(\frac{1}{n}\right)}{\frac{1}{n}} = 1 \neq 0$$

$$\therefore \sum_{n=1}^{\infty} a_n \text{ is divergent (L.C.T)}$$



11.5 Alternating series (A.S.T)



شكل السيريس

$$\sum (-1)^n b_n$$

$$\sum (-1)^{n-1} b_n$$

عند وجود

$$(-1)^n \text{ or } (-1)^{n\pm 1} \text{ or } (-1)^{n\pm 2} \dots$$



خطوات الحل



1) $b_n = (-1)^n$ السيكوينس بدون

2) $\lim_{n \rightarrow \infty} b_n$

= 0

≠ 0

3) b_n is decreasing

$$f'(x) < 0$$

$$b_{n+1} - b_n < 0$$

**Convergent
(A.S.T)**

**Divergent
(N.T.T)**

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} (-1)^n \sin\left(\frac{1}{n}\right)$$

Let $b_n = \sin\left(\frac{1}{n}\right)$

$$\lim_{n \rightarrow \infty} b_n$$

$$\lim_{n \rightarrow \infty} \sin\left(\frac{1}{n}\right) = \sin\left(\frac{1}{\infty}\right) = \sin 0 = 0$$

Let $f(x) = \sin\left(\frac{1}{x}\right)$

$$f'(x) = \cos\left(\frac{1}{x}\right) \left(-\frac{1}{x^2}\right) < 0 \quad \forall x \in [1, \infty)$$

$\Rightarrow \therefore f(x)$ is decreasing $\Rightarrow \therefore b_n$ is decreasing

$\Rightarrow \therefore \sum_{n=1}^{\infty} (-1)^n \sin\left(\frac{1}{n}\right)$ is convergent (A.S.T)



Text book question



Test the series is convergence or divergence

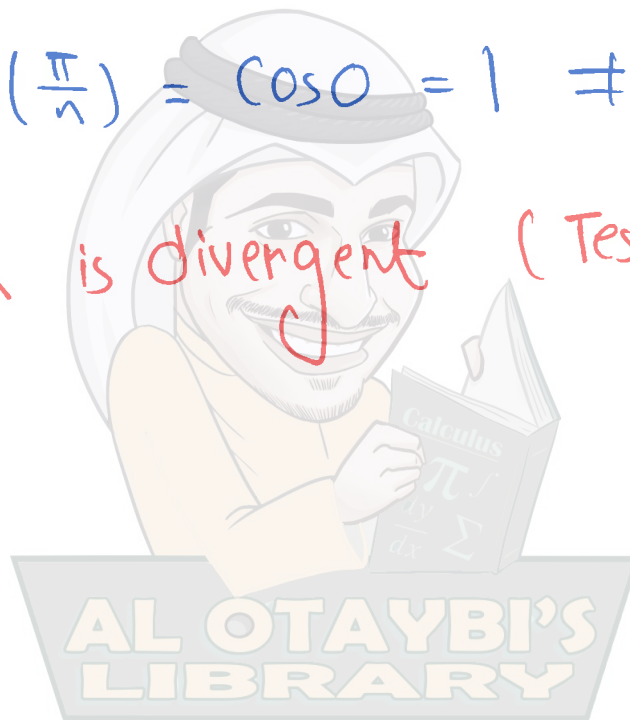
$$\sum_{n=1}^{\infty} (-1)^n \cos\left(\frac{\pi}{n}\right) \rightarrow a_n$$

Let $b_n = \cos\left(\frac{\pi}{n}\right)$

$$\lim_{n \rightarrow \infty} b_n$$

$$\lim_{n \rightarrow \infty} \cos\left(\frac{\pi}{n}\right) = \cos 0 = 1 \neq 0$$

$\therefore \sum_{n=1}^{\infty} a_n$ is divergent (Test of divergence)



Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} (-1)^{n+1} n \sin\left(\frac{1}{n}\right)$$

Let $b_n = n \sin\left(\frac{1}{n}\right)$

$$\lim_{n \rightarrow \infty} b_n$$

$$\lim_{n \rightarrow \infty} n \sin\left(\frac{1}{n}\right) \quad \infty \sin 0$$

$$\lim_{n \rightarrow \infty} \frac{n \sin\left(\frac{1}{n}\right)}{\frac{1}{n}} = 1 \neq 0$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} (-1)^{n+1} n \sin\left(\frac{1}{n}\right)$ is divergent (Test of divergence) N.T.T

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} (-1)^{n+2} \left(\frac{n+4}{n}\right)^n$$

Let $b_n = \left(\frac{n+4}{n}\right)^n$

$$\lim_{n \rightarrow \infty} b_n$$

$$\lim_{n \rightarrow \infty} \left(\frac{n+4}{n}\right)^n$$

$$\lim_{n \rightarrow \infty} \left(\frac{n}{n} + \frac{4}{n}\right)^n = \lim_{n \rightarrow \infty} \left(1 + \frac{4}{n}\right)^n = e^4 \neq 0$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} (-1)^{n+2} \left(\frac{n+4}{n}\right)^n$ is divergent (Test of divergence) N.T.T

Determine whether the series is convergent or divergent

$$\sum_{n=4}^{\infty} (-1)^n \frac{n^2}{\sqrt{e^n}}$$

Let $b_n = \frac{n^2}{\sqrt{e^n}}$

$\lim_{n \rightarrow \infty} b_n$

$\lim_{n \rightarrow \infty} \frac{n^2}{\sqrt{e^n}} = \lim_{n \rightarrow \infty} \frac{n^2}{(e^n)^{\frac{1}{2}}} = \lim_{n \rightarrow \infty} \frac{n^2}{e^{\frac{1}{2}n}} \quad \frac{\infty}{\infty}$

Let $f(x) = \frac{x^2}{e^{\frac{1}{2}x}}$

L'Hop $\lim_{x \rightarrow \infty} \frac{x^2}{e^{\frac{1}{2}x}} = \lim_{x \rightarrow \infty} \frac{2x}{e^{\frac{1}{2}x} \cdot \frac{1}{2}} = \lim_{x \rightarrow \infty} \frac{4x}{e^{\frac{1}{2}x}} \quad \frac{\infty}{\infty}$

L'Hop $\lim_{x \rightarrow \infty} \frac{4}{e^{\frac{1}{2}x} \cdot \frac{1}{2}} = \frac{4}{\infty} = 0$

$f'(x) = \frac{e^{\frac{1}{2}x} \cdot 2x - x^2 e^{\frac{1}{2}x} \cdot \frac{1}{2}}{(e^{\frac{1}{2}x})^2}$
 $= \frac{x e^{\frac{1}{2}x} [2 - \frac{1}{2}x]}{e^x} < 0 \quad \forall x \in (4, \infty)$

$\Rightarrow \therefore f(x)$ is decreasing $\Rightarrow \therefore b_n$ is decreasing

$\Rightarrow \therefore \sum_{n=4}^{\infty} (-1)^n \frac{n^2}{\sqrt{e^n}}$ is convergent (A.S.T)

Show that the following series is convergent

$$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n!}$$

Let $b_n = \frac{1}{n!}$

$$\lim_{n \rightarrow \infty} b_n$$

$$\lim_{n \rightarrow \infty} \frac{1}{n!} = \frac{1}{\infty} = 0$$

$$b_n = \frac{1}{n!} \rightarrow b_{n+1} = \frac{1}{(n+1)!}$$

$$b_{n+1} - b_n = \frac{1}{(n+1)!} - \frac{1}{n!}$$

$$= \frac{1}{(n+1)n!} - \frac{1}{n!}$$

$$= \frac{1}{(n+1)n!} - \frac{n+1}{n!(n+1)}$$

$$= \frac{1 - (n+1)}{(n+1)n!}$$

$$= \frac{\cancel{1} - n - \cancel{1}}{(n+1)n!} = \frac{-n}{(n+1)n!} < 0 \quad \forall \quad n \in [1, \infty)$$

$\Rightarrow \therefore b_n$ is decreasing

$\Rightarrow \therefore \sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n!}$ is Convergent (A.S.T)

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{\cos(n\pi)n^2}{n^2 + 3n + 5} \rightarrow a_n$$

$$\sum_{n=1}^{\infty} \frac{(-1)^n n^2}{n^2 + 3n + 5}$$

$$\text{Let } b_n = \frac{n^2}{n^2 + 3n + 5}$$

$$\lim_{n \rightarrow \infty} b_n$$

$$\lim_{n \rightarrow \infty} \frac{n^2}{n^2 + 3n + 5}$$

$$\lim_{n \rightarrow \infty} \frac{n^2}{n^2 \left[1 + \frac{3}{n} + \frac{5}{n^2} \right]} = \frac{1}{1+0+0} = 1 \neq 0$$

$\therefore \sum_{n=1}^{\infty} a_n$ is divergent (Test of divergence)



Text book question



Test the series is convergence or divergence

$$\frac{1}{\ln 3} - \frac{1}{\ln 4} + \frac{1}{\ln 5} - \frac{1}{\ln 6} + \frac{1}{\ln 7} - \dots$$

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\ln(n+2)} \rightarrow a_n$$

Let $b_n = \frac{1}{\ln(n+2)}$

$$\lim_{n \rightarrow \infty} \frac{1}{\ln(n+2)} = \frac{1}{\infty} = 0$$

Let $f(x) = \frac{1}{\ln(x+2)}$

$$f'(x) = \frac{0 - (1) \cdot \frac{1}{x+2}}{[\ln(x+2)]^2}$$

$$= -\frac{1}{(x+2)[\ln(x+2)]^2} < 0 \quad \forall x \in [1, \infty)$$

$\therefore f(x)$ is decreasing $\longrightarrow \therefore b_n$ is decreasing

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent (A.S.T)

a) Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}$$

b) How many terms we need to find the sum of accuracy

$$(|\text{error}| < 0.01)$$

a) Let $b_n = \frac{1}{n^2}$

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{1}{n^2} = 0$$

$$\text{Let } f(x) = \frac{1}{x^2} = x^{-2}$$

$$f'(x) = -2x^{-3} = -\frac{2}{x^3} < 0 \quad \forall x \in [1, \infty)$$

$\therefore f(x)$ is decreasing on $[1, \infty)$

$\therefore b_n$ is decreasing on $[1, \infty)$

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent (A.S.T)

b) $|a_n| = \frac{1}{n^2}$

$$|a_{n+1}| = \frac{1}{(n+1)^2}$$

$$|a_{n+1}| < \text{error}$$

$$\frac{1}{(n+1)^2} < 0.01$$

$$\frac{1}{(n+1)^2} < \frac{1}{100}$$

$$(n+1)^2 > 100$$

$$\therefore \boxed{n \geq 10}$$

a) Determine whether the series is convergent or divergent

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!}$$

b) How many terms we need to find the sum of accuracy

$$(|error| < 10^{-2})$$

a) Let $b_n = \frac{1}{(2n)!}$

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{1}{(2n)!} = \frac{1}{\infty} = 0$$

$$b_n = \frac{1}{(2n)!} \longrightarrow b_{n+1} = \frac{1}{(2(n+1))!}$$

b) $|a_n| = \frac{1}{(2n)!}$

$$|a_{n+1}| = \frac{1}{(2(n+1))!} = \frac{1}{(2n+2)!}$$

$$|a_{n+1}| < \text{error}$$

$$\frac{1}{(2n+2)!} < 10^{-2}$$

$$\frac{1}{(2n+2)!} < \frac{1}{10^2} \rightarrow \frac{1}{100}$$

$$(2n+2)! > 100$$

$$n > 7/2$$



Multiple choice



By the alternating series estimation theorem the smallest upper bound on the absolute error that result if the sum of the series

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$$

is approximated by the 8th partial sum is

- a) $\frac{1}{9}$ b) $\frac{1}{64}$ c) $\frac{1}{72}$ d) $\frac{1}{81}$ e) $\frac{1}{100}$

Dec 2017

$$|a_n| = \frac{1}{n^2} \rightarrow |a_{n+1}| = \frac{1}{(n+1)^2}$$

n=8
error?

$$|a_{n+1}| > \text{error}$$

$$\frac{1}{(n+1)^2} > \text{error}$$

when n=8

$$\frac{1}{9^2} > \text{error}$$

$$\frac{1}{81} > \text{error}$$

$$\therefore \text{error } \frac{1}{81} \quad \text{(d)}$$

G.T.H.P.I

N.a.B.L

کوکتیل !!

Determine whether the following series is convergent or divergent

$$\sum_{n=1}^{\infty} \left[e^{\frac{1}{n}} - e^{\frac{1}{n+1}} \right]$$

Telescoping

$$\sum_{n=1}^{\infty} \frac{1}{n^5}$$

P-series

$$\sum_{n=3}^{\infty} \frac{2}{n^2 - 2n}$$

اختلاف 11
L.C.T
Telescoping

$$\sum_{n=1}^{\infty} \frac{1}{n+5}$$

L.C.T

$$\sum_{n=1}^{\infty} \frac{2^n}{n^3 + 2}$$

N.T.T
(test of divergence)

$$\sum_{n=1}^{\infty} e^{n+1} 3^{1-n}$$

Geometric Series

$$\sum_{n=1}^{\infty} \frac{\sin^2 \sqrt{n}}{\sqrt[3]{n^4 + 4}}$$

B.C.T



Multiple choice



Select the series that is convergent

a) $\sum_{n=1}^{\infty} [\tan(n) - \tan(n+1)]$ b) $\sum_{n=1}^{\infty} \frac{1}{2n+1}$

c) $\sum_{n=1}^{\infty} \frac{3n}{n+1}$ d) $\sum_{n=1}^{\infty} [\arctan(n) - \arctan(n+1)]$

e) $\sum_{n=1}^{\infty} \pi^n$

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(d)

$$a_n = \tan^{-1}n - \tan^{-1}(n+1)$$

$$S_n = a_1 + a_2 + \dots + a_{n-1} + a_n$$

$$= [\cancel{\tan^{-1}1} - \cancel{\tan^{-1}2}] + [\cancel{\tan^{-1}2} - \cancel{\tan^{-1}3}] + \dots + [\cancel{\tan^{-1}(n-1)} - \cancel{\tan^{-1}(n-1+1)}] + [\cancel{\tan^{-1}n} - \tan^{-1}(n+1)]$$

$$S_n = \tan^{-1}1 - \tan^{-1}(n+1)$$

$$\lim_{n \rightarrow \infty} \left[\frac{\pi}{4} - \tan^{-1}(n+1) \right] = \frac{\pi}{4} - \frac{\pi}{2} = -\frac{\pi}{4}$$

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent (Telescoping)

$$\text{Sum} = -\frac{\pi}{4}$$



Multiple choice



Select the series that is convergent

a) $\sum_{n=1}^{\infty} \frac{1}{n+2}$ b) $\sum_{n=1}^{\infty} \frac{n}{n^2+2}$ c) $\sum_{n=1}^{\infty} \frac{(-1)^n}{2n+1}$

d) $\sum_{n=1}^{\infty} \cos \frac{1}{n}$ e) $\sum_{n=1}^{\infty} \ln n$

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1(c) ✓

Let $b_n = \frac{1}{2n+1}$

$\lim_{n \rightarrow \infty} b_n$

$\lim_{n \rightarrow \infty} \frac{1}{2n+1} = \frac{1}{\infty} = 0$

Let $f(x) = \frac{1}{2x+1}$

$f'(x) = \frac{0 - (1)(2)}{(2x+1)^2} = \frac{-2}{(2x+1)^2} < 0 \quad \forall x \in [1, \infty)$

$\Rightarrow \therefore f(x)$ is decreasing $\Rightarrow \therefore b_n$ is decreasing

$\therefore \sum_{n=1}^{\infty} \frac{(-1)^n}{2n+1}$ is convergent (A.S.T)
(Alternating Series test)



True or False !!



$$\sum_{n=1}^{\infty} n^{-\sin(1)} \quad \text{is convergent}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^{\sin 1}}$$

$$P = \sin 1 < 1$$

divergent (P-series)

⇒ False

$$\sum_{n=1}^{\infty} n^{-\cosh x} \quad \text{is convergent for any real number } x \neq 0$$

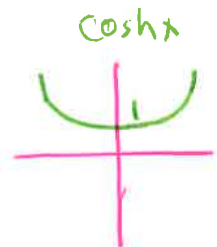
$$\sum_{n=1}^{\infty} \frac{1}{n^{\cosh x}}$$

$$P = \cosh x > 1$$

$$\forall x \neq 0$$

Convergent (P-series)

⇒ True





True or False !!



The series $\sum_{n=3}^{\infty} (n^{-e} - e^{-n})$ is converges

$$\sum_{n=3}^{\infty} n^{-e} - \sum_{n=3}^{\infty} e^{-n}$$

$$\sum_{n=3}^{\infty} \frac{1}{n^e}$$

$$p = e > 1$$

$\therefore \sum_{n=3}^{\infty} \frac{1}{n^e}$ is convergent
(P-series)

$$\sum_{n=3}^{\infty} \left(\frac{1}{e}\right)^n$$

$$r = \frac{1}{e}$$

$$|r| = \frac{1}{e} < 1$$

$\therefore \sum_{n=3}^{\infty} \left(\frac{1}{e}\right)^n$ is convergent
(Geometric Series)

$$\therefore \sum_{n=3}^{\infty} (n^{-e} - e^{-n})$$

is convergent

True

The series $\sum_{n=1}^{\infty} \frac{3^n}{p^n}$ is converges for all

p such that $|p| \geq 3$

$$\sum_{n=1}^{\infty} \left(\frac{3}{p}\right)^n$$

$$r = \frac{3}{p}$$

$\therefore \sum_{n=1}^{\infty} \left(\frac{3}{p}\right)^n$ is Converges

$$\therefore |r| < 1$$

$$\left|\frac{3}{p}\right| < 1$$

$$\frac{|3|}{|p|} < 1$$

$$* |p|$$

$$3 < |p|$$

$$\therefore |p| > 3 \quad (p \neq 3)$$

False